



Novara Materials™

Series A Overview

A production-ready advanced materials platform positioned to scale across mobility, architecture, and consumer heating — backed by 65+ Battelle patents and a \$70M manufacturing facility running today.

S E R I E S A

\$5M Round

\$3.5M remaining

\$40M pre-money

Close: July 2026

June 2026

The Bottom Line

Two structural advantages that no competitor can replicate

MANUFACTURING EDGE

via entrotech partnership

- ▶ Exclusive access to entrotech's \$70M roll-to-roll thin-film production platform
- ▶ Proven, scalable manufacturing from day one — no facility build-out required
- ▶ Turn-key capability compresses time-to-market vs. competitors building from scratch
- ▶ Durable films already deployed on production vehicles

IP PORTFOLIO

via Battelle Memorial Institute — exclusive, perpetual license

- ▶ 65+ patents — covering material compositions, process methods, and device architectures
- ▶ Deep, defensible moat: competitors cannot replicate without infringing this portfolio
- ▶ Active military/defense programs underway through Battelle — classified applications
- ▶ >\$60M of Battelle + USAF R&D investment underlying the IP — you didn't fund it

Novara combines things deep-tech investors almost never get at once — exclusive IP, existing scale manufacturing, and validated customer pull

Manufacturing + IP = An unmatched ability to deliver thin-film heaters at scale

THE TECHNOLOGY

Why Carbon Nanotubes Change Everything

Exceptional thermal properties

CNT films conduct and distribute heat orders of magnitude more efficiently than conventional resistive wire — at a fraction of the weight and thickness

Uniform heat distribution

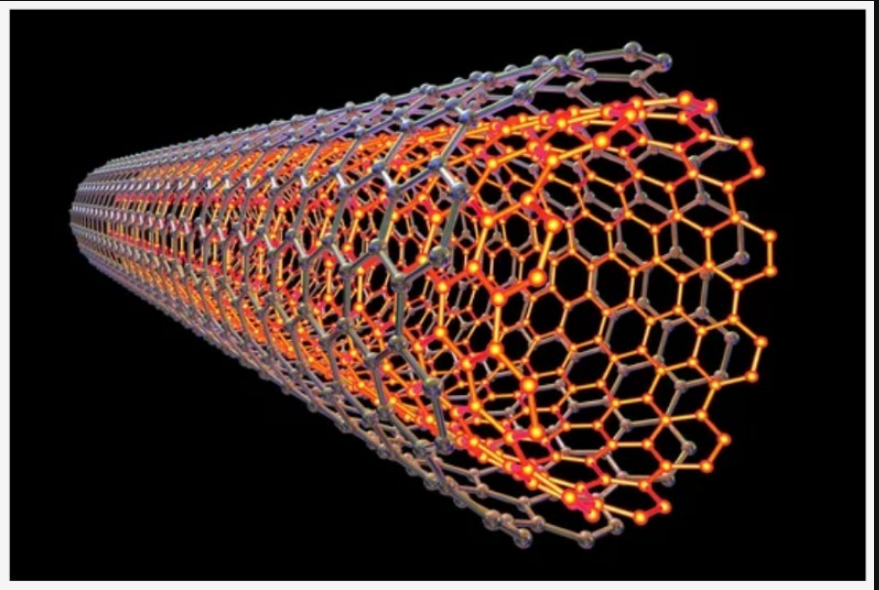
Unlike nichrome wire with its hot/cold spots, CNT coatings disperse heat across the entire film surface — enabling lower maximum operating temperatures

Incredible form-factor and design flexibility

~10 mil thick, ~1.2 oz/ft², operable from 6VDC to 480VAC — can be applied to virtually any substrate, geometry, or application

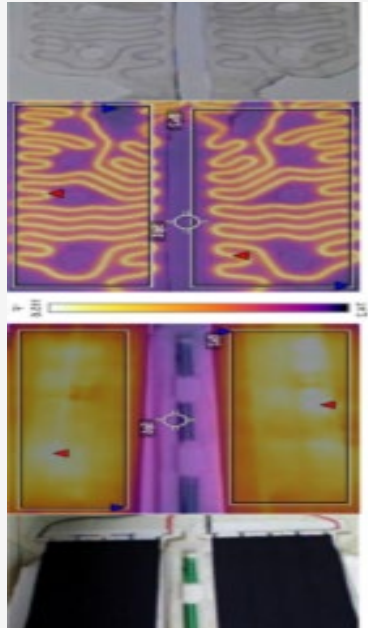
Validated science

Originating from Battelle and >\$60M of Battelle + USAF direct investment into 65+ defensible patents



TECHNOLOGY PROOF

Demonstrated by Hyundai / HATCI OEM Testing



Top: Nichrome Wire (hot/cold spots)
Lower: Novara Panel (uniform heat)

Metric	Nichrome Wire	Novara CNT Film	Improvement
Avg. Temperature	92°F / 33.3°C	100°F / 37.7°C	Higher w/ less energy
Power Required	30 watts	15 watts	50% less power
Time to Temperature	1 min 35 sec	13 seconds	86% faster
Weight	~10× heavier	~1.2 oz/ft ²	90% lighter
Heat Distribution	Hot/cold spots	Uniform ✓	Lower max temp
Price	—	Equal	Same cost, better output

**Application dependent. Data from HATCI seat heating evaluation.*

50%
Less power

90%
Less weight

86%
Faster heat

COMPETITIVE POSITION

No Competing Technology Matches This Stack

Feature	Novara CNT Films	Nichrome Wire	PTC Heaters	Etched Foil
Heat Distribution	✓	—	✗	—
Weight	✓	—	—	✗
Thickness	✓	—	—	✗
Manufacturing	✓	—	✗	✗
Design Flexibility	✓	✗	—	✗
IP Protection	✓	✗	✗	✗

Why Novara Wins

- ✓ 65+ patents — deep IP moat
- ✓ 90% weight savings
- ✓ Uniform heat = lower max temp
- ✓ Commercially viable pricing
- ✓ \$70M facility ready now
- ✓ Customizable form factors

Uniform heat + thin-film flexibility + production-ready scale: no competing technology combines all three

MARKET OPPORTUNITIES

Multiple Markets, One Platform Technology

CNT thin-film heating addresses distinct markets from a single manufacturing base

CONSUMER / FOODSERVICE

NOW — Sampling

- Delivery bags
- Flameless chafers
- Outdoor furniture

First revenue H2 2026

ARCHITECTURE

2–3 YRS

- Snow/ice melt — KOSU in process
- FAA PEGASAS Project
- Hangar under-slab radiant heat

MOBILITY

3–4 YRS

- BorgWarner — active PoC
- EV battery warming
- Seat / passenger cabin microclimate

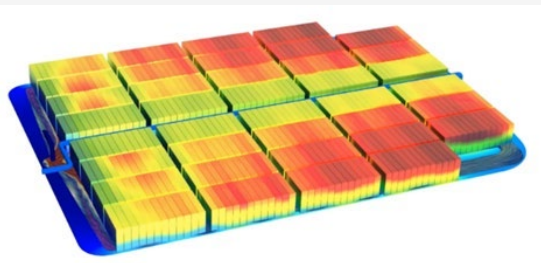
AEROSPACE / ENERGY

5+ YRS

- Honeywell Aerospace — SOW received
- GE Aerospace — in pipeline
 - Critical surface de-icing
 - Drone & fixed-wing leading edges
 - Wind turbine blade protection (GE Vernova)

MOBILITY — DEMONSTRATED TRACTION

Demonstrated Effective with Mobility OEMs



EV battery pack thermal simulation

BATTERY WARMING — HATCI / HYUNDAI DATA

Key Metric	Baseline (existing coolant)	Surface Heating (measured results)	Improvement (Computer Modeled)
Charge Time 10→80%	74 min	28 min	-62%
Warm-Up (-20°C→20°C)	58 min	11 min	-81%
Energy Required	3.8 kWh	1.7 kWh	-56%
EV Range	312 km	388 km	+24%

IN CABIN PASSENGER COMFORT — DEMONSTRATED DURABILITY

28,000 ingress/egress cycles using 95th-percentile male robot | **80%** power savings vs. whole-cabin HVAC | **-40°C to 120°C** operating range | Chemical resistance to gasoline, diesel, kerosene, salt fog

+24%

EV range extension

-81%

Faster battery warm-up

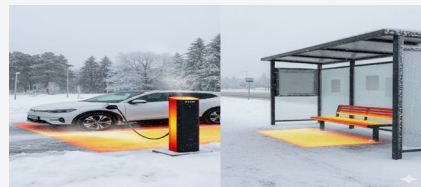
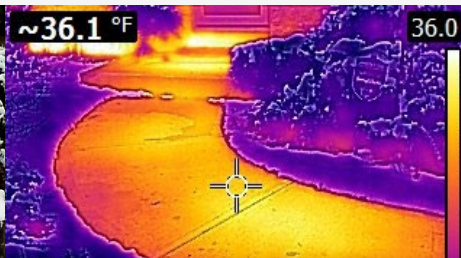
5 yrs

Hyundai validation data

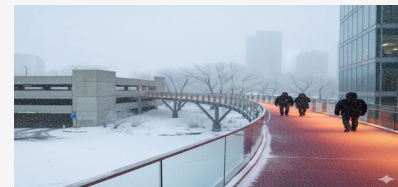
PoC

BorgWarner PoC active

Real-World Installation. Year 5 of Operation.



EV charging stations · Bus stops



Heated walkways

Residential — Year 5

Proven long-term durability

- ▶ Exterior walkway installation active since 2020
- ▶ FLIR-validated uniform heat distribution
- ▶ KOSU project underway to finalize commercial products

Gemini Field Trial @KOSU

Aug–Oct 2026 install window

- ▶ Concrete retrofit and under-epoxy formats
- ▶ ECS / Equity Construction Solutions partnership
- ▶ Commercial building and hangar applications
- ▶ Q1 2027 performance data expected

Advanced Purchase Order: June 2026

- ▶ 10,000 square feet of snow/ice melt
- ▶ 5,000 square feet of interior radiant
- ▶ 12 additional field trial sites ready
- ▶ Target August install

Signed LOI for Agriculture: Target install October 2026

CONSUMER — NEAR-TERM REVENUE

First Revenue Products — H2 2026

Low-heat CNT films cut to size — manufacturing capacity ready, active sampling across four categories

MANUFACTURING CAPACITY ADVANTAGE

Nichrome today: **400K units/yr @ ~\$35 cost** →
Novara target: **6M units/yr @ <\$10 cost** (15× capacity, ¼ cost)

Food Delivery Bags

Sampling active

Pizza chains, gig driver market,
DoorDash/Uber Eats channel

Outdoor Furniture

Sampling active

Patio furniture OEMs, Hotels, Venues

Flameless Chafing Dishes

Sampling active

Hospitality, catering, foodservice — replaces
gel fuel heating

Personal Care

In exploration

Heating pads, blankets, therapeutic products

Revenue path

1. Q3–Q4 2026: Convert samples to first POs
2. 2H 2026: Initial revenue, 1–2 categories
3. Q4 2026: Expand to 3–4 channels
4. 2027: Recurring revenue base established
5. Gross margins maintained at 50%+ at entrotech cost structure

Staggered Resourcing/Effort follows the clock

Each vertical's resource curve, positioned against its horizon to revenue

Why we run five verticals at once — and why that is necessary, not ‘unfocused’.

Resource allocation and intensity is matched to each vertical's clock.

Near-term revenues fund long-term moats.



Agriculture

Chef's Garden pilot



Architecture

Snow-melt/ floor heat



Consumer

Delivery bags/ chafers



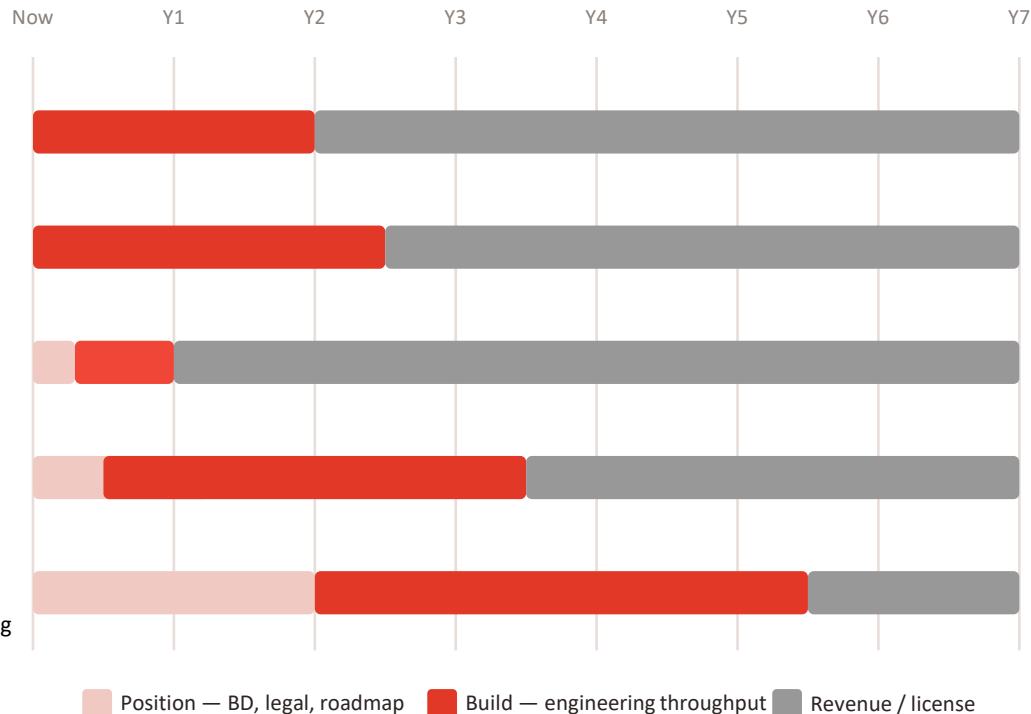
Mobility

BorgWarner / Gentex



Aerospace

Honeywell / GE de-icing



Long-horizon verticals need presence now, throughput later — i.e. start late on aerospace and we may never play.

Our Unfair Advantage: The entrotech Platform

A \$70M world-class Columbus, OH facility competitors cannot replicate — already running, already scaling

Exclusive access — competitors cannot license the same capability

Roll-to-roll: clean, eco-friendly, fraction of the energy vs. nichrome wire manufacturing



entrotech \$70M clean room facility — Columbus, OH

\$70M

Facility value

60"

Film width

110 ft/min

Speed

25 yrs

entrotech track record

\$1.5B

Revenue history

MARKET SIGNALS

Opportunity Pipeline Exceeds Capacity

Validated opportunities across four verticals — capital required to capture in parallel

ACTIVE DEVELOPMENT

Snow/Ice Melt — Concrete Retrofit

KOSU testing · ECS partnership · Field trial pipeline Aug 2026

Airplane Hangar Radiant Heat

Under-slab & under-epoxy · OSU Airport Testing Resumes Summer 2026

Foodservice Products

Delivery bags & flameless chafing · Revenue target H2 2026

BorgWarner Battery Heating

Intercell eCooling ribbon · Active PoC underway

NEW INBOUND INQUIRIES

Honeywell Aerospace

Critical surface de-icing: drones & fixed-wing — builds on prior USAF work

GE Aerospace / GE Vernova

Same as Honeywell + Wind Turbines

NASA + VISTA

StarLab space station at KOSU's new G.W. Carver Science Park

More inbound demand than our current team can serve
· Capital enables parallel pursuit of each project

LEADERSHIP

World-Class Board. Experienced Operators.

Kent Helfrich

Chairman, Novara Board

- Former CTO — General Motors
- Former VP, GM R&D
- Former President, GM Ventures

Dr. Amy Heintz

Technical Fellow, Battelle

- Inventor of HeatCoat technology
- Battelle Distinguished Inventor
- Battelle Inventor of the Year

Matt McFarland

VP Technology, Battelle

- Decades of commercialization experience
- Former VP Commercialization, Nationwide Children's Hospital

Gary Booth

Board Member

- 35-year career at Procter & Gamble
- Founder/Board Member, multiple venture funds & startups

Core Operating Team

Andy Dickson

President

25+ yrs early-stage tech · 15 yrs nanotechnology · OSU Finance/CS

Renee Albarano

CFO

20+ yrs corporate finance · Deep start-up operations expertise

Brian Lipp

Dir. Product Dev & Ops

25+ yrs product dev · Former Battelle researcher & founder

Brian Gamble

Dir. Business Development

Leading mobility & EV market development · Key account mgmt

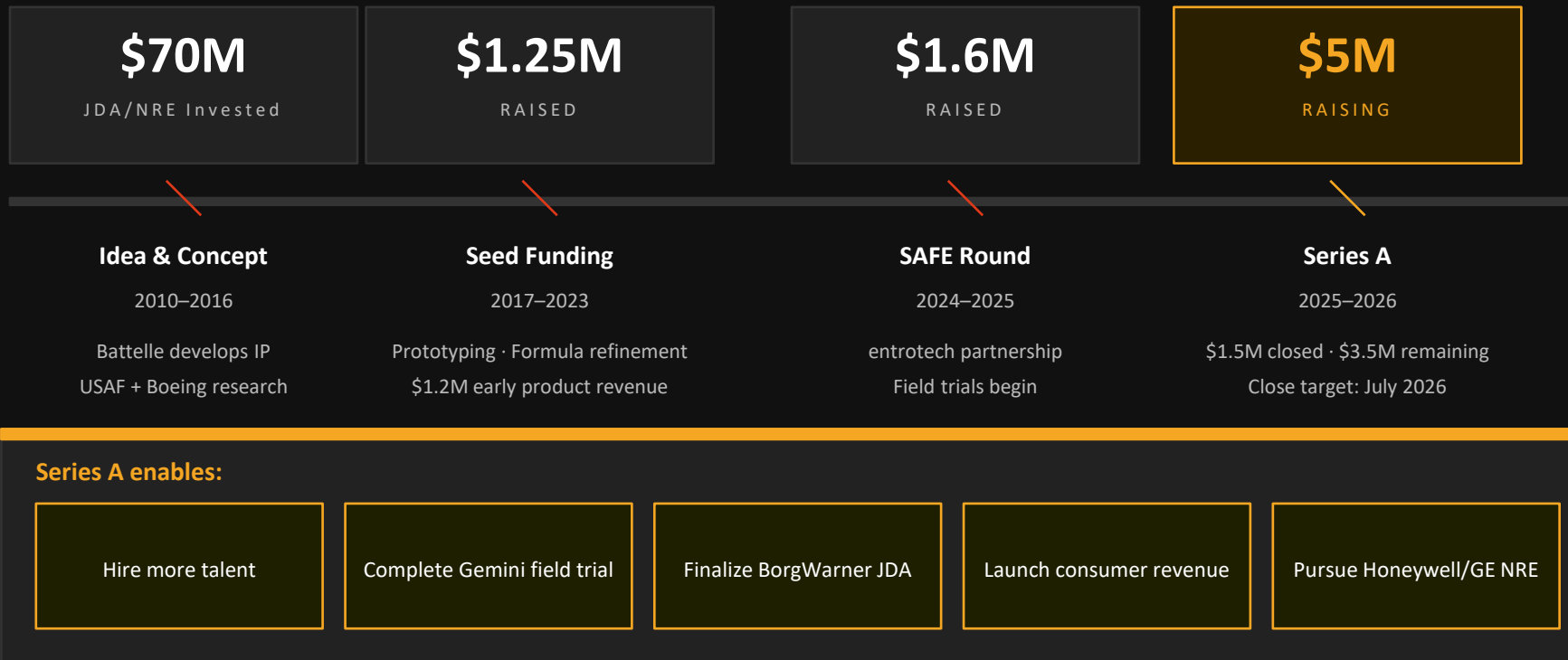
Andrew Strange*

Technical Dir. (entrotech)

30+ yrs R&D · 20+ US patents in coatings/film · 20 yrs entrotech R&D

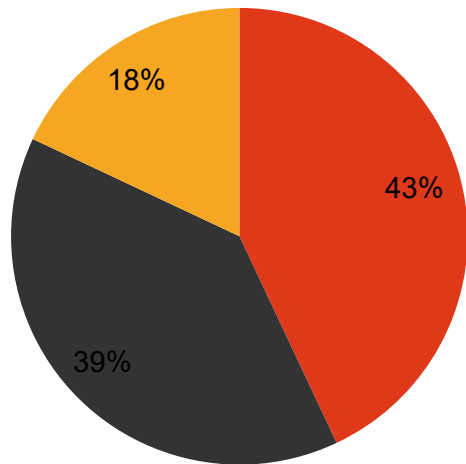
INVESTMENT HISTORY

De-Risking is Done. Now Ready to Scale.



USE OF FUNDS

\$5M Deployed Across People, Partners, and Products



- Operations & Development (43%)
- Talent (39%)
- Product Testing (18%)

12-MONTH MILESTONES

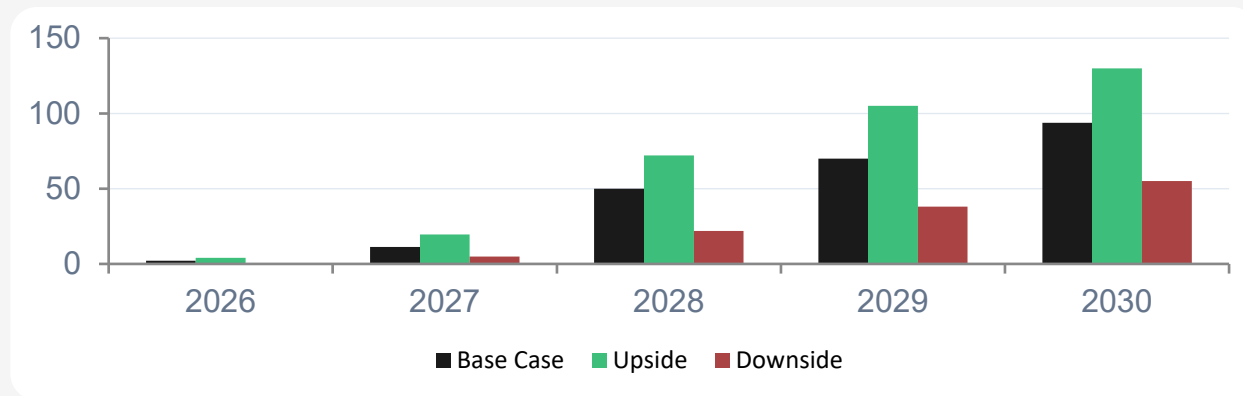
1. Q2 2026: First consumer revenue (delivery bags or chafing dishes)
2. Q3 2026: Gemini field trial installation complete
3. Q4 2026: BorgWarner JDA finalized
4. Q1 2027: Gemini performance data, architecture commercial rollout
5. Q2 2027: Cash flow positive (base case)

Without this capital, Novara cannot pursue Architecture and Consumer simultaneously — forcing a sequential path that delays revenue by 12–18 months.

FINANCIAL MODEL

Revenue Scenarios — 2026 through 2030

Scenario	Key Assumption	2026	2027	2028	2030	Breakeven
Base Case	Consumer H2 2026 · 1 mobility JDA · Gemini data Q1 2027	\$2.2M	\$11.3M	\$49.9M	\$93.7M	Q2 2027
Upside	2 mobility JDAs + aerospace NRE + consumer scale	\$4.0M	\$19.5M	\$72M	\$130M+	Q4 2026
Downside	Consumer only through 2026, mobility delayed 12 months	\$0.5M	\$4.8M	\$22M	\$55M	Q3 2027



Model notes

- Consumer GM 50%+
- 1 mobility JDA = \$500K–\$1.5M
- Architecture: 30M ft² customer-informed estimate
- No Battelle military revenue modeled
- Full model in data room

SERIES A — THE ASK

\$3.5M to convert proven technical advantage into flagship commercial scale

Join investors who recognize that the manufacturing, IP, and OEM traction are already in place.

Total Series A	\$5,000,000
Closed to Date	\$1,500,000
Remaining	\$3,500,000
Pre-Money Valuation	\$40,000,000
Liquidation Preference	1× non-participating
Subsidiary Option	Invest in specific vertical
Target Close	July 2026

Exit scenarios: Tier-1 auto supplier acquisition (Mobility) · Architecture material co. · Aerospace prime · Strategic license per vertical

INVESTMENT THESIS

Why Invest Now

1 Production-Ready Manufacturing

\$70M turn-key facility running today. No capital build required. Ready to scale when needed.

2 Owned IP Portfolio

65+ patents 35+ trade secrets, exclusive perpetual license from Battelle.
>\$60M of R&D you didn't fund. Deep moat.

3 Proven OEM Traction

OEM engagement has proven viability.
Active PoC with BorgWarner. Honeywell inbound. Legitimate demand pipeline.

4 Supply Chain Dynamics

Insensitive to volatility in silver/copper prices;
immune to tariffs
Many N.A. OEMs prefer/require N.A. suppliers

5 Capital-Efficient Model

Leveraged Battelle R&D + entrotech manufacturing. Capital goes to commercialization, not infrastructure.

6 Multiple Exit Paths

Strategic exits or licenses per vertical: Tier-1 auto, architecture material companies, aerospace/defense primes.



Andy Dickson, President

andy@novaramaterials.com · Data room available under NDA

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This presentation contains forward-looking statements. Actual results may differ materially from projections.